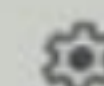
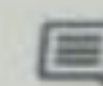


Commands + Code + Text

```
plt.title('Customer Churn')  
plt.show()
```





```
[17] # Define the tools and technologies used in the project
tools_and_technologies = {
    "Programming Language": "Python",
    "Machine Learning Library": "Scikit-learn",
    "Data Visualization Library": "Matplotlib",
    "Data Analysis Library": "Pandas",
    "Model Deployment": "Flask or Django"
}

# Print the tools and technologies used in the project
print("Tools and Technologies:")
for tool, technology in tools_and_technologies.items():
    print(f"{tool}: {technology}")

Tools and Technologies:
Programming Language: Python
Machine Learning Library: Scikit-learn
Data Visualization Library: Matplotlib
Data Analysis Library: Pandas
Model Deployment: Flask or Django

# Import necessary libraries
import matplotlib.pyplot as plt

# Define the data
ages = [25, 30, 35, 20, 40, 22, 38, 45, 28, 32]
usage = [100, 200, 300, 400, 500, 150, 350, 600, 250, 450]
churn = [0, 0, 1, 0, 1, 0, 1, 1, 0, 1]

# Create a scatter plot
plt.figure(figsize=(10, 6))
plt.scatter(ages, usage, c=["red" if c == 1 else "green" for c in churn])
plt.xlabel("Age")
plt.ylabel("Usage")
plt.title("Customer Churn")
plt.show()
```




```
[14] print(f"Mean: {sum(usage) / len(usage)}")
print(f"Min: {min(usage)}")
print(f"Max: {max(usage)}")
```

Summary statistics for age:

Mean: 30.0

Min: 20

Max: 40

Summary statistics for usage:

Mean: 300.0

Min: 100

Max: 500

Define the dataset

```
customers = [
    {"customer_id": 1, "age": 25, "usage": 100},
    {"customer_id": 2, "age": 30, "usage": 200},
    {"customer_id": 3, "age": 35, "usage": 300},
    {"customer_id": 4, "age": 20, "usage": 400},
    {"customer_id": 5, "age": 40, "usage": 500}
]
```

Create a new feature: usage per age

```
for customer in customers:
    customer["usage_per_age"] = customer["usage"] / customer["age"]
```

Print the updated dataset

```
for customer in customers:
    print(customer)
```

```
{'customer_id': 1, 'age': 25, 'usage': 100, 'usage_per_age': 4.0}
{'customer_id': 2, 'age': 30, 'usage': 200, 'usage_per_age': 6.666666666666667}
{'customer_id': 3, 'age': 35, 'usage': 300, 'usage_per_age': 8.571428571428571}
{'customer_id': 4, 'age': 20, 'usage': 400, 'usage_per_age': 20.0}
{'customer_id': 5, 'age': 40, 'usage': 500, 'usage_per_age': 12.5}
```

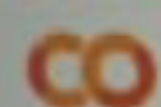
[17] Define the tools and technologies used in the project

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Search





```
[12] print("Model built and trained successfully.")
```

```
Model built and trained successfully.
```

```
[13] # Predict the target variable
      predictions = model.predict(X)

      # Print the predictions
      print(predictions)
```

```
[0 0 1 0 1]
```

```
# Define the dataset
customers = [
    {"customer_id": 1, "age": 25, "gender": "male", "usage": 100},
    {"customer_id": 2, "age": 30, "gender": "female", "usage": 200},
    {"customer_id": 3, "age": 35, "gender": "male", "usage": 300},
    {"customer_id": 4, "age": 20, "gender": "female", "usage": 400},
    {"customer_id": 5, "age": 40, "gender": "male", "usage": 500}
]
```

```
# Calculate summary statistics
ages = [customer["age"] for customer in customers]
usage = [customer["usage"] for customer in customers]
```

```
print("Summary statistics for age:")
print(f"Mean: {sum(ages) / len(ages)}")
print(f"Min: {min(ages)}")
print(f"Max: {max(ages)}")
```

```
print("\nSummary statistics for usage:")
print(f"Mean: {sum(usage) / len(usage)}")
print(f"Min: {min(usage)}")
print(f"Max: {max(usage)}")
```



Commands + Code + Text

[10]

Print the dataset

```
for customer in customers:
    print(customer)
```

```
{'customer_id': 1, 'age': 25, 'gender': 'male', 'usage': 100, 'churn': 0}
{'customer_id': 2, 'age': 30, 'gender': 'female', 'usage': 200, 'churn': 0}
{'customer_id': 3, 'age': 35, 'gender': 'male', 'usage': 300, 'churn': 1}
{'customer_id': 4, 'age': 20, 'gender': 'female', 'usage': 400, 'churn': 0}
{'customer_id': 5, 'age': 40, 'gender': 'male', 'usage': 500, 'churn': 1}
```

Encode categorical variables

```
for customer in customers:
    customer["gender"] = 0 if customer["gender"] == "male" else 1
```

Print the preprocessed dataset

```
for customer in customers:
    print(customer)
```

```
{'customer_id': 1, 'age': 25, 'gender': 0, 'usage': 100, 'churn': 0}
{'customer_id': 2, 'age': 30, 'gender': 1, 'usage': 200, 'churn': 0}
{'customer_id': 3, 'age': 35, 'gender': 0, 'usage': 300, 'churn': 1}
{'customer_id': 4, 'age': 20, 'gender': 1, 'usage': 400, 'churn': 0}
{'customer_id': 5, 'age': 40, 'gender': 0, 'usage': 500, 'churn': 1}
```

[12] from sklearn.linear_model import LogisticRegression

Define the features and target variable

```
X = [[customer["age"], customer["gender"], customer["usage"]] for customer in customers]
y = [customer["churn"] for customer in customers]
```

Create a logistic regression model

```
model = LogisticRegression()
```

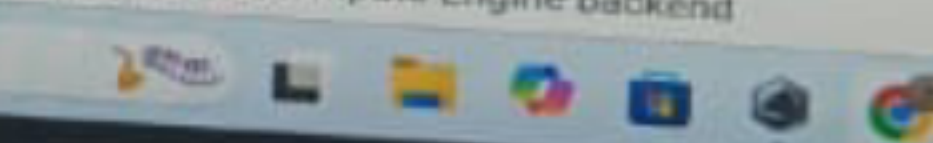
Train the model

```
model.fit(X, y)
```

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Search





Coefficients: [[0.61682541 0.03081422]]

[7] #Define a new customer

new_customer = [[32, 250]]

Predict whether the new customer will churn or not

prediction = model.predict(new_customer)

Print the prediction

print("Prediction:", prediction)

Prediction: [0]

Define the project objectives

```
project_objectives = ["Develop a model that can accurately predict customer churn.",  
                      "Identify the key factors that influence customer churn.",  
                      "Improve customer retention by targeting high-risk customers."]
print(project_objectives)
```

['Develop a model that can accurately predict customer churn.', 'Identify the key factors that influence customer churn.', 'Improve customer retention by targeting high-risk customers.']

[10] customers = [

```
{ "customer_id": 1, "age": 25, "gender": "male", "usage": 100, "churn": 0 },  
{ "customer_id": 2, "age": 30, "gender": "female", "usage": 200, "churn": 0 },  
{ "customer_id": 3, "age": 35, "gender": "male", "usage": 300, "churn": 1 },  
{ "customer_id": 4, "age": 20, "gender": "female", "usage": 400, "churn": 0 },  
{ "customer_id": 5, "age": 40, "gender": "male", "usage": 500, "churn": 1 }
```

Print the dataset

```
for customer in customers:  
    print(customer)
```





```

▶ customers = [
    {"age": 25, "usage": 100, "churn": 0},
    {"age": 30, "usage": 200, "churn": 0},
    {"age": 35, "usage": 300, "churn": 1},
    {"age": 20, "usage": 400, "churn": 0},
    {"age": 40, "usage": 500, "churn": 1}
]

# Define the features and target variable
X = [[customer["age"], customer["usage"]] for customer in customers]
y = [customer["churn"] for customer in customers]

# Print the features and target variable
print("Features:", X)
print("Target Variable:", y)

```

↕ Features: [[25, 100], [30, 200], [35, 300], [20, 400], [40, 500]]
Target Variable: [0, 0, 1, 0, 1]

```

▶ from sklearn.linear_model import LogisticRegression

# Create a logistic regression model
model = LogisticRegression()

# Train the model
model.fit(X, y)

# Print the model's coefficients
print("Coefficients:", model.coef_)

```

↕ Coefficients: [[0.61682541 0.03081422]]

[] #Define a new customer

... Initializing Python 3 Google Compute Engine backend